COMSM0118: Foundations of Cyber Security assessments Group Assessment

Module Title:	Foundations of Cyber Security
Level:	MSc
Assessment title:	Threat Modelling (group work: 40%)
Indicative Deadline:	12:00, Friday 03 October 2025
Assessed intended learning outcomes:	• Demonstrate research skills exploring the vulnerabilities and threats in a real-world system and communicate this diagrammatically and in writing. • Demonstrate the ability the perform a threat assessment of an arbitrary system and their knowledge of STRIDE and attack trees. • Demonstrate knowledge of risk and threat mitigation. • Learn to work in groups. • Learn to communicate about complex technical topics in a manner that is suitable for a wider audience.
Weighting within module:	This assessment is worth 40% of the overall module mark.
Task details and instructions:	This assessment asks you to work in your allocated groups to undertake a detailed threat assessment of a system. Your task will be to research the system in groups and build a data flow diagram. From your data flow diagram, we would like you to perform a STRIDE[1] analysis and develop attack trees showing how individual components of the system could be compromised and leveraged into an attack against the operation or safety of the system. You will be asked to suggest countermeasures against the threats you identify. **Hyundai Santa Fe Case Study** Modern cars help their passengers to stay safe by automating part of the driving process. These driver assistance systems can help drivers stay within speed limits, warn when attention is diminished and when a collision may be imminent. These cars also, however, include cloud-based network features and smart assistants. The interplay between these cloud-based tools and the safety-critical systems within the car opens up new attack vectors for hackers. For this assessment we will focus on the Hyundai Santa Fe—a large car produced by Hyundai since 2001. Recent versions of the car feature the following safety systems: • a blind-spot detection system with safe-exit feature; • a rear-view camera with cross-traffic detection, rear collision avoidance and smart boot hatch; • a smart cruise control with forward collision avoidance; • a lane assist system with driver attention warning.

[1] Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service and Elevation of Privilege.

	Your task is to **research one of these systems** and its attack surface; starting from the information in the Hyundai Santa Fe Quick Reference Guide and Owner's Manual. These are provided to you as part of the assessment pack. A summary of the relevant features and pointers to relevant sections in the car manuals is provided in a document. This is to make it easier for you to identify relevant parts of the manuals and documentation.
	Acknowledgments This assessment is based on the Driver Assistance Safety System & Security Case Study by Bastian Tenbergen, produced as part of the Case Studies in Support of CyBOK 1.0 Selected Topics.

Deliverables:	For your submission, you are asked to provide a report of a maximum of **3,000 words (excluding additional figures and references)**. In other words, additional figures and references are not included in the word count and will not contribute to the 3,000-word limit. Number of words should appear on the cover page. Your report should contain the following: • A high-level description of the system you are modelling including any assumptions about the system and its resources that you are making. (Weighting: 10%). • A data-flow diagram showing how information from sensors and surrounding systems interacts with the system you have chosen to model. (Weighting: 20%). • Using your data-flow diagram as a starting point, perform a threat assessment of the system using the STRIDE methodology highlighting threats to the system and showing which of the STRIDE threats may be applicable in this context. ***Three threats*** should be highlighted and discussed in detail. Please note that attaching a generated report from the Microsoft Threat Modelling tool is not an acceptable answer. But using the tool is highly recommended. The description of the threats in the report must be your own. (Weighting: 30%). • For each of the ***three threats*** you highlight, produce an attack tree illustrating the security and safety properties that the attack would violate and further attacks it might enable, as well as the compromises that would need to succeed for the threat to transform into a successful attack. (Weighting: 20%). • For each of the ***three threats*** highlighted suggest a series of mitigations to reduce the threat. (Weighting: 10%). A further 10% of the available marks will be allocated to writing style, grammar, coherence of the argument and typography. The use of references to improve your discussion and illustrate threats and their mitigations where applicable are encouraged. **Each group member must also submit a reflective learning log (max. 1 side of A4 but flexible) discussing their learning and contributions as well as contributions of their group members and group dynamics. The minimum font size should be 11 pt font, Arial or Helvetica.**

Groups

Group 1	**Office hours (30min)**	**Teaching Assistant**
Amanda Ayumi Alcantara	Tuesday and Thursday	Andy Baldrian (andy.baldrian@bristol.ac.uk) & Marvin
Boyang Wang		
Huiqi Liu		
Ryan Yang		
Yongtao Shi		
Group 2		
Jie Yu Huang	Tuesday and Thursday	Alyah Al Fageh (alyah.alfageh@bristol.ac.uk) & Marvin
Sanika Pravin Mahamulkar		
Tailai Liu		
Tianju Yang		
Yi Li		
Zhuo Chen		
Group 3		
Shobhit Jha	Tuesday and Thursday	Lucy Davies (lucy.davies@bristol.ac.uk) & Marvin
Abdi Abdi Hakim Hared		
Xiao Liu		
Yidan Xia		
Yuang Gao		
Group 4		
Zimu Hu	Tuesday and Thursday	Cheng Cheng (cheng.cheng@bristol.ac.uk) & Marvin
Ashwin Kotharamath		
Esperanza Li		
Xinqi Lu		
Xiao Chen		
Group 5		
Amit Kiran Deb	Tuesday and Thursday	Zhiyuan Xu (zhiyuan.xu@bristol.ac.uk) & Cheng Cheng (cheng.cheng@bristol.ac.uk) & Marvin
Dong Huang		
Tiancheng Liu		
Yuanshuo Tian		
Hongyi Pan		
Group 6		
Shrivatsa Shridhar Deshpande	Tuesday and Thursday	Alyah Al Fageh (alyah.alfageh@bristol.ac.uk) & Marvin
Wanlan Qiu		
Churui Liu		
Xiji Zhang		
Sam Cole		
Group 7		
Alexander Vranic	Tuesday and Thursday	Andy Baldrian (andy.baldrian@bristol.ac.uk) & Marvin
Sabir Al Habsi		
Aleena Jolly		
Jerry Ren		
Xuenuoyang Zong		

Kaijia Yang		
Group 8		
Hima Pranavi Kannepogu Jamal Naser Sarwary Xin Shi Xiyan Chen Jerry Wu Yongxing Wang	Tuesday and Thursday	Lucy Davies (lucy.davies@bristol.ac.uk) & Cheng Cheng (cheng.cheng@bristol.ac.uk) & Marvin

Mark Sheet:

Title of Report	**Threat Modelling**		
Group/Pair		**Level M**	**Weighting – 40%**
Unit Director	Kopo Marvin Ramokapane	**Second Marker**	Partha Das Chowdhury

The **Unit Director and the second marker** independently assess project reports and fill in the **Report Comments Form**. Then they input marks on the 0-100 scale into the table on this Mark Sheet. Any significant discrepancies should be carefully discussed.

Report Marking Scheme:

Report marker's name	**Description (10%)**	**Data-flow diagram (20%)**	**Threat Assessment (30%)**	**Attack Trees (20%)**	**Mitigations (10%)**	**Persuasiveness (10%)**	**Total mark (weighted average)**
Marvin							
Partha							
	Report mark (agreed marks of the two markers):						

Mark awarded (See University Generic Marking Criteria)	

Marking Criteria:

Grade	**0-20 scale**	**0-100 scale**	**Criteria to be satisfied**
A	20 19 18	100 94 89	• Work would be worthy of dissemination under appropriate conditions • Mastery of advanced methods and techniques at a level beyond that explicitly taught • Ability to synthesise and employ in an original way ideas from across the subject • In group work, there is evidence of an outstanding individual contribution • Excellent presentation • Outstanding command of critical analysis and judgement
	17 16 15	83 78 72	• Excellent range and depth of attainment of intended learning outcomes • Mastery of a wide range of methods and techniques • Evidence of study and originality clearly beyond the bounds of what has been taught • In group work, there is evidence of an excellent individual contribution • Excellent presentation
B	14 13 12	68 65 62	• Attained all the intended learning outcomes • Able to use well a range of methods and techniques to come to conclusions • Evidence of study, comprehension and synthesis beyond the bounds of what has been explicitly taught • Very good presentation of material • Able to employ critical analysis and judgement • Where group work is involved there is evidence of a productive individual contribution
C	11 10 9	58 55 52	• Some limitations in attainment of learning objectives, but has managed to grasp most of them • Able to use most of the methods and techniques taught • Evidence of study and comprehension of what has been taught • Adequate presentation of material • Some grasp of issues and concepts underlying the techniques and material taught • Where group work is involved there is evidence of a positive individual contribution
D	8 7	48 45	• Limited attainment of intended learning outcomes • Able to use a proportion of the basic methods and techniques taught • Evidence of study and comprehension of what has been taught, but grasp insecure • Poorly presented • Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
E	6	42	
	5	35	• Attainment of only a minority of the learning outcomes • Able to demonstrate a clear but limited use of some of the basic methods and techniques taught • Weak and incomplete grasp of what has been taught • Deficient understanding of the issues and concepts underlying the techniques and material taught

	1-4	7-29	• Attainment of nearly all the intended learning outcomes deficient • Lack of ability to use at all or the right methods and techniques taught Inadequately and incoherently presented • Wholly deficient grasp of what has been taught • Lack of understanding of the issues and concepts underlying the techniques and material taught
F	0	0	• No significant assessable material, absent or assessment missing a “must pass” component